

From Quantum Many-Body Scars to Slow Dynamics

Exact Diagonalization of a Deformed PXP Chain

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Abstract

This note summarizes a numerical study of a periodically closed, constrained spin chain derived from the PXP model. The Hamiltonian combines the standard Rydberg-blockaded spin-flip term with staggered one-spin and three-spin diagonal perturbations. Exact diagonalization is used to resolve the spectrum, identify eigenstates with anomalously large overlap with the charge-density-wave product state $|\mathbb{Z}_2\rangle$, and compute equilibrium thermodynamic quantities. The largest-overlap scarred eigenstate is found in the low-energy part of the finite-size spectrum. The nonequilibrium calculation probes a connected local X -autocorrelation after a thermal quench. Its early-time decay is fitted directly, without selecting peaks, by linear regression of $\log C_X(t)$ against $\log t$. The observed power-law decay provides a concrete measure of slow relaxation. It motivates comparison with Griffiths-like dynamics, but it does not by itself establish a Griffiths phase.

1 Motivation

Experiments on Rydberg-atom arrays revealed persistent oscillations following a quench from an ordered state, despite the absence of conventional integrability. The corresponding constrained spin model supports atypical eigenstates embedded in an otherwise thermal spectrum. These states, now called quantum many-body scars, violate the usual expectation that all finite-energy-density eigenstates of a generic interacting system locally resemble thermal equilibrium.

The present project asks how this scarred spectral structure changes when the PXP chain is deformed so that the most strongly scarred state moves toward low energy. The numerical workflow connects three complementary diagnostics: the exact energy spectrum and $\mathbb{Z}_2$ overlaps, the canonical heat capacity, and the relaxation of a local autocorrelation function. The first diagnoses atypical eigenstates, the second tests their thermodynamic context, and the third measures the timescale on which local memory is lost.

2 Constrained spin model

Consider a periodic chain of L spin-1/2 degrees of freedom. Let X_i and Z_i denote Pauli operators, and define

$$n_i = \frac{1 + Z_i}{2}, \quad P_i = 1 - n_i = \frac{1 - Z_i}{2}. \quad (1)$$

The Rydberg blockade restricts the Hilbert space to configurations satisfying

$$n_i n_{i+1} = 0 \quad \text{for every } i, \quad (2)$$

with site labels understood modulo L . Within this constrained space, a local spin flip is equivalent to

$$\tilde{X}_i = P_{i-1} X_i P_{i+1}, \quad (3)$$

and the standard PXP Hamiltonian is

$$H_{\text{PXP}} = \sum_{i=0}^{L-1} \tilde{X}_i. \quad (4)$$

The model studied here adds staggered diagonal perturbations,

$$H(g, r) = H_{\text{PXP}} + g \sum_{i=0}^{L-1} (-1)^i (Z_i + r Z_{i-1} Z_i Z_{i+1}). \quad (5)$$

The parameter g sets the overall strength of the deformation and r fixes the relative weight of the three-spin term. The staggering reduces one-site translation symmetry to translation by two sites. Equation (5) is implemented in a custom constrained basis and diagonalized with QuSpin.

The reference product state is

$$|\mathbb{Z}_2\rangle = |1010 \dots\rangle, \quad (6)$$

which is invariant under translation by two sites and therefore belongs to the zero-momentum sector of that symmetry.

3 Exact diagonalization and scarred eigenstates

For the spectral calculation, the Hamiltonian is represented in the two-site translation sector $k = 0$. The current calculation uses $L = 20$ and obtains the full dense eigendecomposition

$$H(g, r) |E_n\rangle = E_n |E_n\rangle. \quad (7)$$

For each eigenstate, the principal scar diagnostic is its overlap with the ordered state,

$$w_n = |\langle \mathbb{Z}_2 | E_n \rangle|^2. \quad (8)$$

The spectral location of the initial state can also be summarized by

$$E_{\mathbb{Z}_2} = \langle \mathbb{Z}_2 | H | \mathbb{Z}_2 \rangle = \sum_n E_n w_n. \quad (9)$$

The numerical spectrum contains a small set of eigenstates whose values of w_n are anomalously large compared with nearby eigenstates. These outliers are identified as quantum many-body scar candidates. For the deformation of principal interest, $g = -0.4$ and $r = 0.2$, the eigenstate with the largest $\mathbb{Z}_2$ overlap lies in the low-energy region of the finite-size spectrum. This placement differs from the familiar picture of scarred states near the middle of the spectrum and makes low-temperature thermodynamic diagnostics especially relevant.

The half-chain entanglement entropy of every eigenstate is evaluated after lifting the states from the symmetry-reduced basis to the full constrained basis. It is used as an additional color-coded spectral diagnostic. Unfolded nearest-neighbor level spacings are computed from the middle two thirds of the spectrum, where edge effects are weaker. These quantities help distinguish isolated nonthermal states from a global loss of ergodicity, although a systematic finite-size scaling analysis is required before drawing a thermodynamic-limit conclusion.

Figure 1 collects the spectral diagnostics and the corresponding quench dynamics. The upper-left panel makes the anomalously large $\mathbb{Z}_2$ overlaps visible and places the strongest-overlap states toward the low-energy side of the spectrum. The lower panels show that the same initial state produces long-lived, damped oscillations in both a local observable and the return probability.

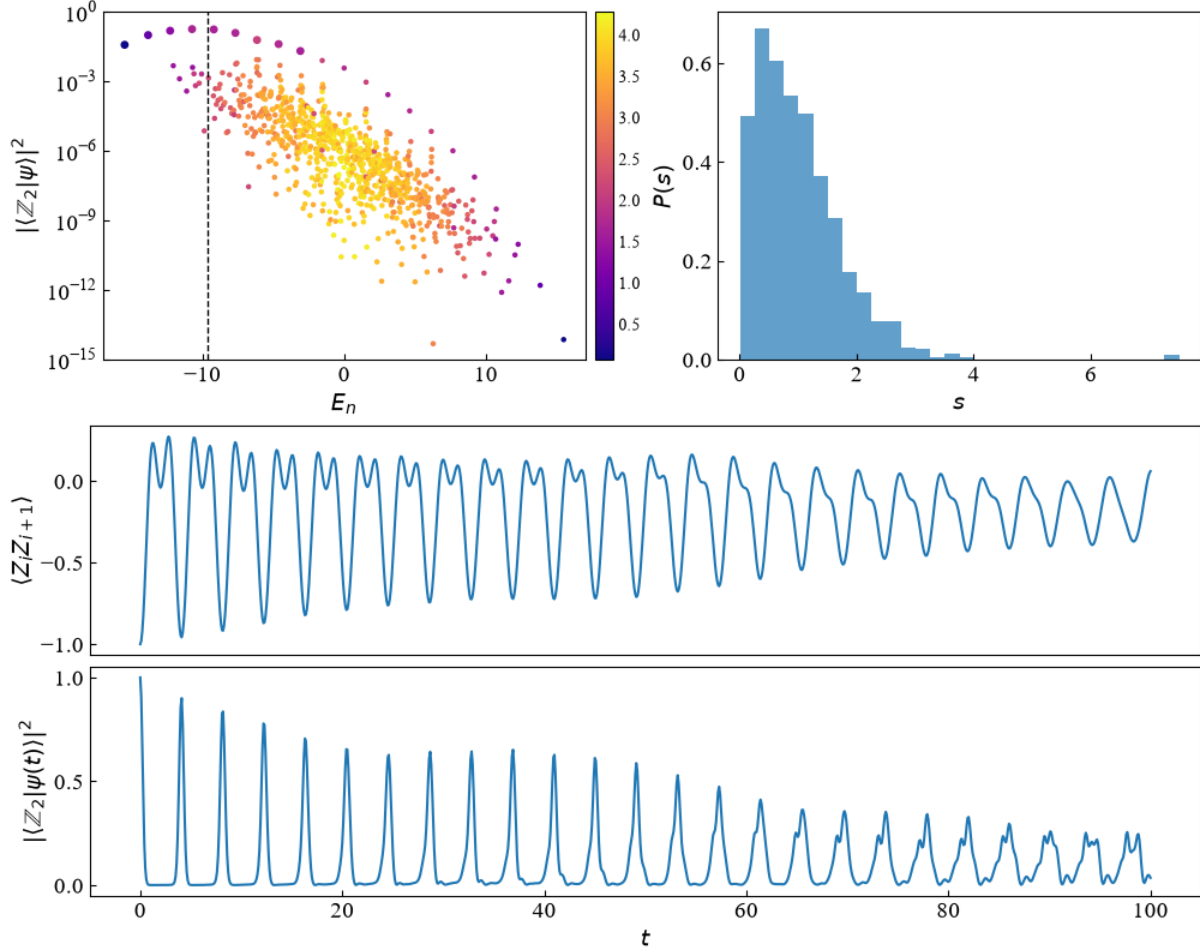


Figure 1: Spectral and dynamical diagnostics for $L = 20$, $g = -0.4$, and $r = 0.2$. Upper left: overlap $|\langle \mathbb{Z}_2 | E_n \rangle|^2$ versus energy; the color scale gives the half-chain entanglement entropy, and the dashed line marks $E_{\mathbb{Z}_2}$. Upper right: distribution of unfolded nearest-neighbor level spacings. Lower panels: the site-averaged nearest-neighbor correlation $L^{-1} \sum_i \langle Z_i Z_{i+1} \rangle$ and the return probability $|\langle \mathbb{Z}_2 | \psi(t) \rangle|^2$ after a quench from $|\mathbb{Z}_2\rangle$.

4 Canonical energy and heat capacity

The thermodynamic calculation uses the eigenvalues from all two-site momentum sectors, rather than only the $k = 0$ sector used for the scar plot. If $E_{n,k}$ denotes the spectrum in an independent momentum block, the partition function is reconstructed as

$$Z(\beta) = \sum_{k \in \mathcal{K}_{\text{ind}}} m_k \sum_n e^{-\beta E_{n,k}}, \quad (10)$$

where $m_k = 2$ for a conjugate pair k and $-k$, while self-conjugate sectors have $m_k = 1$. In practice, the global ground-state energy is subtracted inside the Boltzmann factors to avoid numerical

underflow; this shift is restored in the mean energy.

With $k_B = 1$, the canonical internal energy and heat capacity are

$$U(T) = \langle H \rangle_\beta = \frac{1}{Z} \sum_{k,n} m_k E_{n,k} e^{-\beta E_{n,k}}, \quad (11)$$

$$C_V(T) = \frac{\partial U}{\partial T} = \beta^2 \left(\langle H^2 \rangle_\beta - \langle H \rangle_\beta^2 \right). \quad (12)$$

The temperature scan in Figure 2 uses $L = 20$, $r = 0.2$, $0.02 \leq T \leq 2$, and $g = 0, -0.1, \dots, -0.5$. The internal energy rises smoothly with temperature. Each heat-capacity curve has a broad maximum, whose position shifts to higher temperature as the deformation becomes stronger. These maxima show where thermal fluctuations sample the largest concentration of spectral weight.

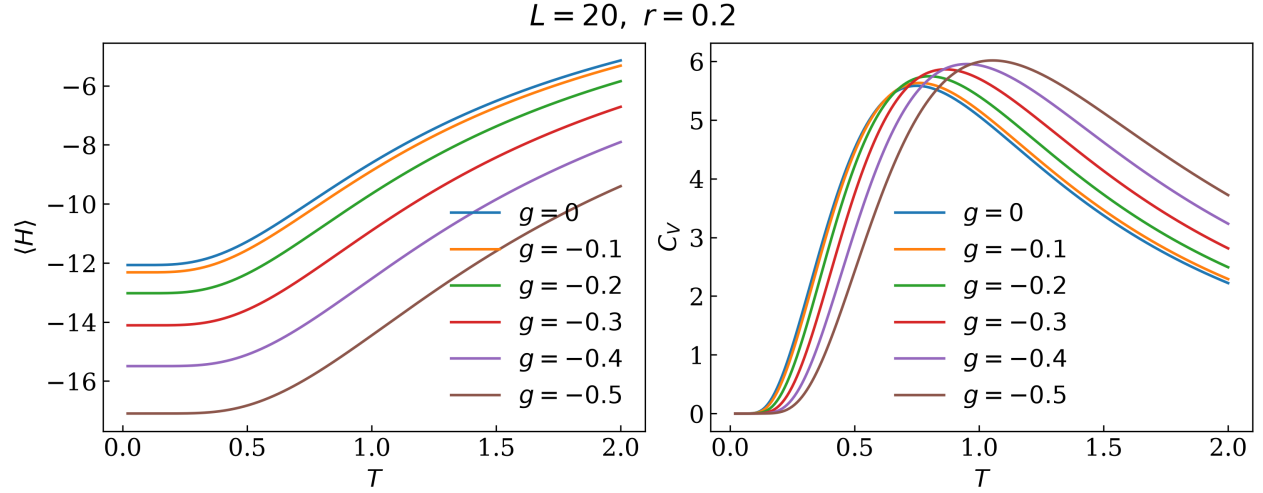


Figure 2: Canonical thermodynamics as a function of temperature for $L = 20$ and $r = 0.2$. The left panel shows the total internal energy $U = \langle H \rangle$; the right panel shows the total heat capacity C_V . Each curve corresponds to a different deformation strength g .

Figure 3 presents the complementary scan in which temperature is fixed and g varies from -0.5 to 0 . At the lowest temperatures, the internal energy follows the deformation of the low-energy spectrum while C_V remains close to zero. At higher temperatures, the heat capacity becomes sensitive to g across the full interval. Taken together, Figures 2 and 3 show how the same spectral rearrangement appears along two cuts through the (g, T) plane. A broad maximum or shoulder in C_V is a crossover indicator at these finite sizes, not by itself evidence of a phase transition.

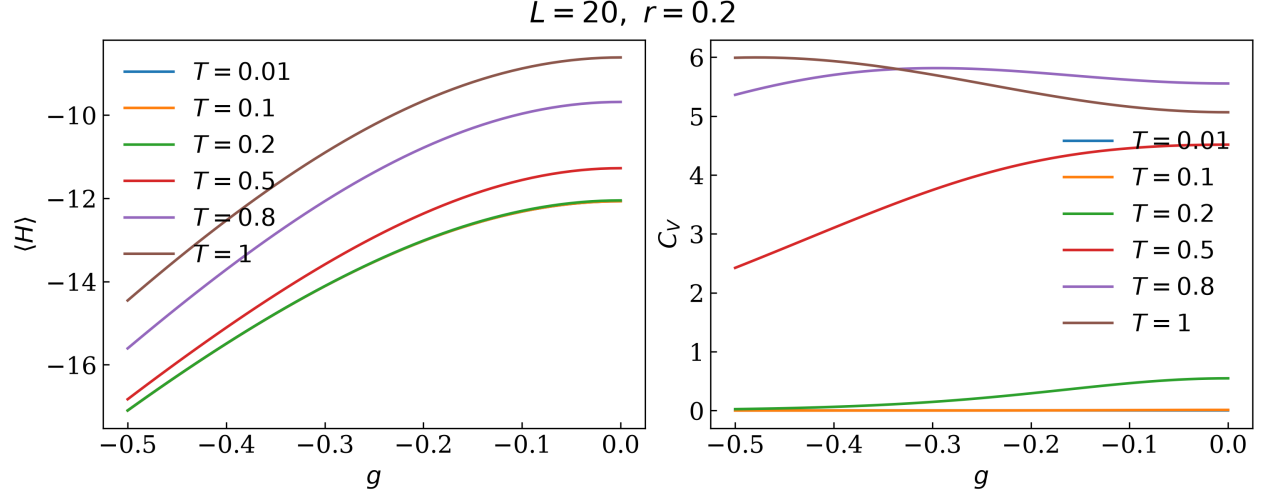


Figure 3: Canonical thermodynamics as a function of deformation strength for $L = 20$ and $r = 0.2$. The left and right panels show $U(g)$ and $C_V(g)$, respectively, at fixed temperatures $T = 0.01, 0.1, 0.2, 0.5, 0.8$, and 1.

5 Connected local autocorrelation

The dynamical calculation starts from a canonical state of the undeformed PXP Hamiltonian,

$$\rho_0 = \frac{e^{-\beta H_{\text{PXP}}}}{\text{Tr}(e^{-\beta H_{\text{PXP}}})}, \quad (13)$$

and evolves operators with the deformed Hamiltonian $H_1 = H(g, r)$,

$$\tilde{X}_i(t) = e^{iH_1 t} \tilde{X}_i e^{-iH_1 t}. \quad (14)$$

The measured quantity is the site-averaged connected autocorrelation

$$C_X(t) = \frac{1}{L} \sum_{i=0}^{L-1} \text{Re} \left[\text{Tr}(\rho_0 \tilde{X}_i(t) \tilde{X}_i) - \text{Tr}(\rho_0 \tilde{X}_i(t)) \text{Tr}(\rho_0 \tilde{X}_i) \right]. \quad (15)$$

The subtraction removes the disconnected one-point contribution. Because ρ_0 need not commute with H_1 , the time-dependent one-point function is retained in that subtraction. The code evaluates Eq. (15) in the eigenbasis of H_1 , allowing all requested times to be obtained from precomputed matrix contractions and phase factors.

The current parameter set is $L = 16$, $g = -0.4$, $r = 0.2$, and $T = 1$. Times are sampled geometrically over $0.1 \leq t \leq 100$, which provides uniform resolution on a logarithmic time axis.

5.1 Power-law fit

The observed decay is modeled by

$$C_X(t) = A t^{-\alpha}. \quad (16)$$

For positive correlation values, Eq. (16) becomes the linear relation

$$\log C_X(t) = \log A - \alpha \log t. \quad (17)$$

Ordinary least-squares regression is applied directly to all correlation samples in the early-time interval

$$0.1 \leq t \leq t_{\text{fit}}, \quad t_{\text{fit}} = 50. \quad (18)$$

No peak detection or upper-envelope selection enters the fit. The fitted exponent is minus the regression slope, and the amplitude is the exponential of the intercept. The plot uses logarithmic axes and marks the fitted time interval with a translucent shaded region without an edge line.

Figure 4 shows the correlation data and the fitted power law. The regression gives $\alpha = 0.486$ for this parameter set. Oscillations remain around the fitted trend, but the dashed line captures the overall decrease across the shaded time window. Data beyond t_{fit} are displayed for comparison and do not enter the regression.

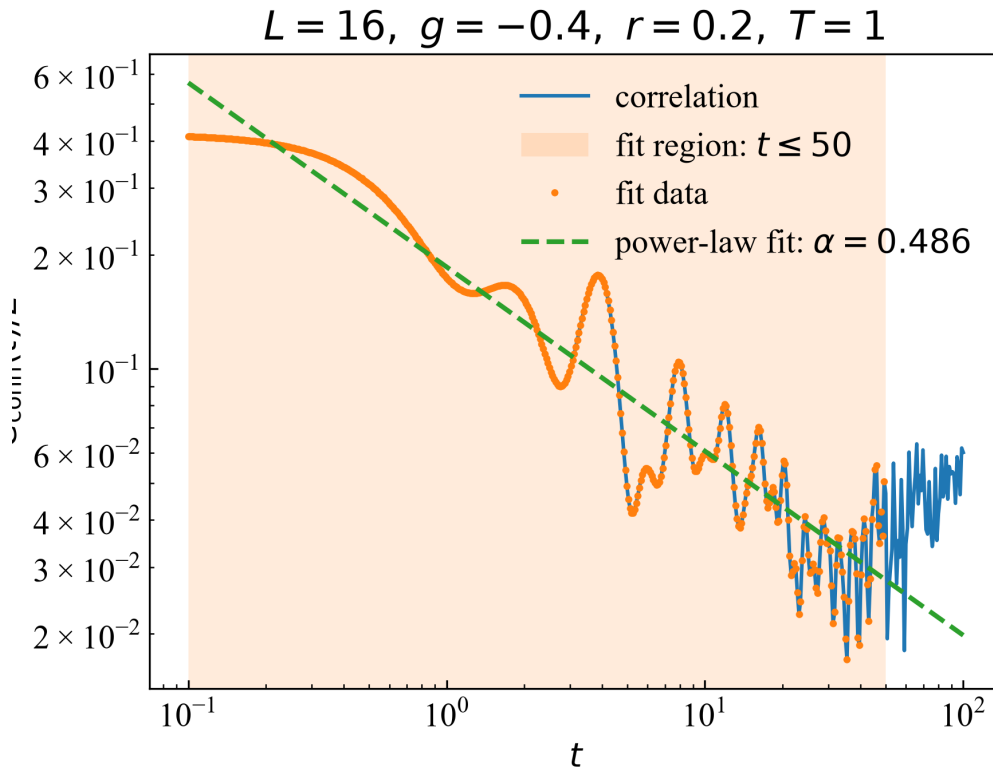


Figure 4: Connected local X -autocorrelation for $L = 16$, $g = -0.4$, $r = 0.2$, and $T = 1$ on logarithmic axes. Orange points inside the shaded region are the data used in the regression over $0.1 \leq t \leq 50$. The green dashed line is the fitted power law $C_X(t) = At^{-\alpha}$ with $\alpha = 0.486$; data at later times are excluded from the fit.

The approximately linear behavior in the log-log representation shows that the finite-size autocorrelation decays as a power law over the selected window. Such slow relaxation is consistent with the broad phenomenology often associated with Griffiths regimes near localization transitions. The present model, however, is clean and staggered rather than spatially random. Power-law decay over a finite time window is therefore a motivation for further tests, not a sufficient identification of a Griffiths phase. Useful next steps include varying L , the fit window, temperature, and deformation parameters; comparing alternative observables; and testing whether the fitted exponent approaches a stable large-system value.

6 Summary

This study follows a single chain of evidence from spectral structure to equilibrium and nonequilibrium observables. Exact diagonalization reveals scarred eigenstates with anomalously large $\mathbb{Z}_2$ overlap and places the strongest-overlap state in the low-energy part of the finite-size spectrum. A symmetry-resolved reconstruction of the full partition function yields the internal energy and heat capacity. Finally, the connected local X -autocorrelation displays a power-law decay when the early-time data are fitted directly on logarithmic axes. Together, these calculations provide a controlled finite-size framework for studying how scar physics may coexist with, or evolve into, slow many-body relaxation.